

# Beyond Expected Goals: A Probabilistic Framework for Shot Occurrences in Soccer

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# What Does It Mean to “Deserve” a Win?

- **Post-match analysis:** We often hear “this team didn’t deserve that result.” But what does this really mean?
- **Intuition:** Deserving refers to the difference in *goal-scoring danger* created or conceded by the two teams.
- **Mainstream approach:** Compare each team’s expected goals (xG)!
- **Our question:** Does xG really capture “deserving”?

# Example 1: Comparing Chances

- **Match:** Manchester City vs. Real Madrid (Feb 19, 2025)
- **Key question:** Which chance is more likely to produce a goal?

## Example 2: Rebounded Shots

- **Match:** Orlando City vs. Philadelphia Union (Feb 22, 2025)
- **Key question:** How does xG measure the value of this sequence?

## Example 2: Rebounded Shots

| Time  | Player  | Shot Outcome (xG)   |
|-------|---------|---------------------|
| 78:01 | Brekalo | Shot Blocked (0.05) |
| 78:02 | Muriel  | Shot Post (0.52)    |
| 78:04 | Pasalic | Shot Post (0.68)    |
| 78:05 | Pasalic | Shot Goal (0.38)    |

**Table:** Four shots in one possession totaling 1.63 xG, but only one goal is possible.

## Example 3: Mbappé's Brilliance

- **Match:** Manchester City vs. Real Madrid (Feb 19, 2025)
- **Key questions:** When is xG credited? What does it value?

# What Are Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context
  - Distance, angle, shot type, pressure, player positions, etc.
  - Often fitted using ML models (e.g., XGBoost) on historical shot data
- **Purposes:**
  - Shot quality estimation
  - Team performance summaries
  - Finishing or goalkeeping “skill” via residuals (GOxG)

# The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.
- So xG answers the following question:

$$\mathbb{P}(\text{goal} \mid \text{shot})$$

- But as we saw, many dangerous moments are *also* about:

$$\mathbb{P}(\text{shot})$$

- What are the consequences of ignoring this?
  - No-shot danger gets 0 credit (Example 1)
  - Rebound sequences inflate summed xG (Example 2)
  - The skill of shot *creation* is under-credited (Example 3)



# Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:
  - ➊ **Will a shot happen?** (creating a chance)
  - ➋ **If it happens, how likely is it to be a goal?** (shot quality)
- **We should model both.**
  - $xS$ : the probability a shot occurs in the next second
  - $xG$ : the probability of a goal given a shot occurs
- Putting them together produces  **$xG+$** : the probability of scoring in the next second.

# Introducing Our New Metric: xG+

- At each time point  $t$ , define:

$$\begin{aligned} \text{xG+}_t &= \mathbb{P}_t(\text{goal scored}) \\ &= \mathbb{P}_t(\text{goal} \mid \text{shot}) \cdot \mathbb{P}_t(\text{shot}) \\ &= \text{xG}_t \cdot \text{xS}_t \end{aligned}$$

- Possession-level aggregation (over  $n$  time points):**

$$\text{xG+}_{\text{poss}} = 1 - \prod_{t=1}^n (1 - \text{xG+}_t)$$

- What this gives us: the probability of scoring (at most once) in the possession.
- Respects the fundamental limit of at most one goal per possession.

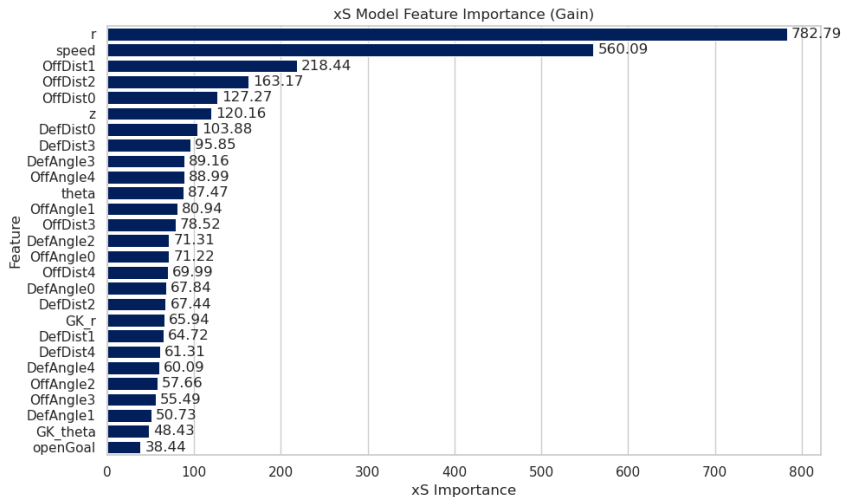
- **Source:** Gradient Sports tracking and event data (EPL 2022–2025)
- **Resolution:** 30 frames per second
- **Inputs:**
  - Player positions (x,y), ball position (x,y,z)
  - Possession indicators, shot events and outcomes
  - Player/team IDs

- Keep frames where the ball is in play and a team has *clear possession* in the attacking third
- Define attacking sequences:
  - Start: possession gained in attacking third
  - End: the defense regains possession or the ball exits the attacking third
- Interpolate ball positions to fill in missing frames

- **Ball details:** distance to goal, angle, height, speed
- **Local pressure:** nearest attackers and defenders relative to ball
- **GK context:** keeper location relative to goal
- **Goal openness (openGoal):**
  - Model defenders as circles (diameter 0.75 m)
  - Compute fraction of goalmouth left unobstructed by defenders

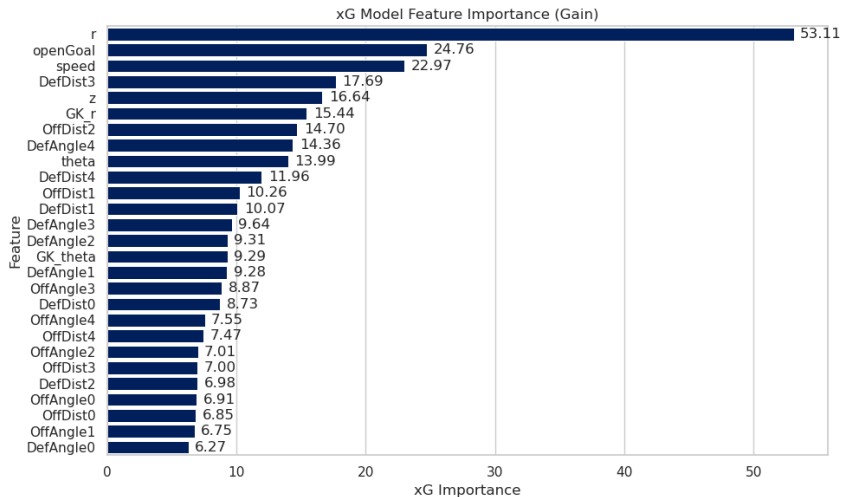
- We train two supervised models using the same feature set:
  - **xS**:  $\mathbb{P}(\text{shot in the next second})$
  - **xG**:  $\mathbb{P}(\text{goal} \mid \text{shot})$
- Fit using 5-fold cross-validated XGBoost
- Evaluated with log loss (lower is better)

# What Drives Shot Creation (xS)?



- Distance and ball speed dominate: proximity and tempo create chances.

# What Drives Shot Quality (xG)?



- Distance, openGoal, and ball speed primarily determine shot quality.



# Does $xG+$ Predict Goals Better than $xG$ ?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!
- **Data:** 3 EPL seasons, 114 matchdays.
- **Out-of-sample test:**
  - 1 Hold out one matchday at a time.
  - 2 Train our models on the other 113 matchdays.
  - 3 Predict goals on the held-out matchday.
- **Compare:**  $xG$  vs  $xS$  vs  $xG+$  (with two possession aggregation rules).

## Moment-level metrics:

- **xS:** Probability of a shot in the next second
- **xG:** Probability of scoring given a shot
- **xG+:**  $xS \times xG$  (probability of scoring in the next second)

## Possession aggregation:

- **At-least-one-per-possession:**  $1 - \prod(1 - p)$
- **Max-per-possession:**  $\max(p)$
- **Baseline:** sum of shot-by-shot xG

# Team-Level Results: $xG+ > xS > xG$

**Table:** Mean Squared Error (MSE) by Metric and Aggregation Method

| Aggregation Method          | $xG+$ | $xS$ | $xG$ |
|-----------------------------|-------|------|------|
| At-least-one-per-possession | 2.84  | 2.90 | 2.94 |
| Max-per-possession          | 2.84  | 2.87 | 2.91 |
| Sum-of-shots                |       |      | 2.90 |

- **Takeaway:** Both  $xG+$  and  $xS$  dominate  $xG$ , with  $xG+$  as the best overall.

# What About Player Ability?

- **Question:** Do we learn more about players from *shot-taking* or *goal-scoring*?
- **Approach:**
  - 1 Aggregate each metric by player and season ( $xG$ ,  $xS$ ,  $xG+$ )
  - 2 Compare to observed goals and shots ( $GOxG$ ,  $SOxS$ ,  $GOxG+$ )
  - 3 Divide by matches played to get per-game metrics
  - 4 Evaluate year-over-year stability (repeatability)

# Key Finding: Finishing Is Not Stable; Shot-Taking Is

**Table:** Year-to-Year Correlation (Stability) of Each Metric (Per Game)

| GOxG | SOxS | GOxG+ |
|------|------|-------|
| 0.12 | 0.63 | 0.35  |

- **Interpretation:**

- **Finishing is very noisy:** GOxG is weakly correlated year-to-year (0.12).
- **Shot-taking is repeatable:** SOxS is strongly correlated (0.63).
- **Takeaway:** Top players are consistently great because they *create more shots than expected*, not because they reliably “finish hot.”

# Top Overperforming Players

xG+ Over Expected (Per Game)

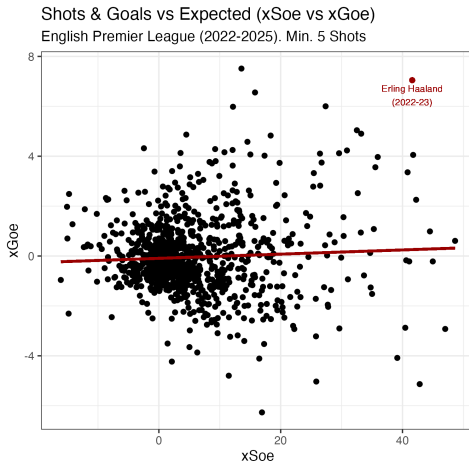
| Season  | Player         | xG+ OE | Goals | Games | Chances |
|---------|----------------|--------|-------|-------|---------|
| 2022-23 | Erling Haaland | 0.52   | 36    | 35    | 444     |
| 2023-24 | Erling Haaland | 0.44   | 30    | 31    | 513     |
| 2024-25 | Omar Marmoush  | 0.41   | 10    | 16    | 293     |
| 2024-25 | Yoane Wissa    | 0.35   | 22    | 35    | 429     |
| 2024-25 | Mohamed Salah  | 0.33   | 27    | 38    | 1042    |
| 2024-25 | Chris Wood     | 0.31   | 23    | 36    | 330     |
| 2023-24 | Cole Palmer    | 0.30   | 19    | 34    | 693     |
| 2024-25 | Alexander Isak | 0.30   | 25    | 34    | 558     |
| 2023-24 | Alexander Isak | 0.30   | 19    | 28    | 351     |
| 2022-23 | Harry Kane     | 0.29   | 25    | 38    | 589     |

xG Over Expected (Per Game)

| Season  | Player          | xG OE | Goals | Games | Chances |
|---------|-----------------|-------|-------|-------|---------|
| 2024-25 | Omar Marmoush   | 0.41  | 10    | 16    | 293     |
| 2024-25 | Chris Wood      | 0.31  | 23    | 36    | 330     |
| 2024-25 | Michael Keane   | 0.21  | 3     | 10    | 19      |
| 2022-23 | Erling Haaland  | 0.52  | 36    | 35    | 444     |
| 2022-23 | Roberto Firmino | 0.25  | 10    | 21    | 253     |
| 2022-23 | Martin Ødegaard | 0.22  | 15    | 37    | 942     |
| 2023-24 | Heung-min Son   | 0.28  | 20    | 35    | 751     |
| 2022-23 | Matias Viña     | 0.26  | 3     | 10    | 66      |
| 2022-23 | Alexander Isak  | 0.24  | 11    | 22    | 346     |
| 2023-24 | Taiwo Awoniyi   | 0.26  | 7     | 19    | 125     |

Note: Minimum 10 Games with a Chance

# Example: Haaland's Record Season



- **Key insight:** Required both taking more shots than expected and favorable finishing variance

- **Best metric for team prediction:**  $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing
- **Why this matters:** match analysis, player evaluation, recruitment, and forecasting
- **Future work:**
  - Sequence models for  $xS$  (point processes / hazard models)
  - Hierarchical team/player effects to control for selection bias
  - Value player actions (shoot vs pass vs carry) with counterfactual  $xG+$
  - Validate in other leagues and deploy in real time



# Questions?



Link to paper, code, and slides

# Acknowledgements

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